

FIRST APP-CONTROLLED BIONIC HAND

A bionic arm that goes by the name of i-Limb

Ultra Revolution is hailed as the first-of-its-kind to be controlled via the comfort of a smartphone app, with the tap of a button. Great way to give muscle power a break
<http://dgit.in/ZqdPYT>

WINDOWS IS DEAD

Following an IDC report on the worst sales period

for the PC segment, a point of view that pegs the blame fairly on Windows 8 and its lacklustre appeal among consumers. But is Windows as we know it truly dead?
<http://dgit.in/15ePLfx>

FOLLOWERS: LOOKING BEYOND THE NUMBERS

What was once the gold standard for measuring your

success on social media – a.k.a your follower count on Twitter, Facebook, Instagram and others – the internet is now closely scrutinizing the quality of your following
<http://dgit.in/Z2AE5i>

80S TECH DEFINES TODAY'S DIGITAL AGE

You'd be surprised just how many of today's gadgets

and gizmos date back to tech breakthroughs achieved in the 1980s. Sift through this list and see if you possess any of these 'tech marvels'
<http://dgit.in/1lfesli>

Human Genome Project is 10 years old, what's next?

- John Hewitt



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This past Sunday marked the 10th anniversary of the sequencing of the human genome. Now that a little time has passed, it makes sense to take stock of what the project has done for us. For the average person, probably not a whole lot. Having an example DNA sequence on file for analysis has slowly trickled out blips of insight for those who look at things like, for example, subtle varieties of various functional genes, or tracing the remnant repeats of viruses that have integrated into the sequence over time. Of particular interest has been how many times they have copied themselves, and spread throughout the genome. However, the main area that sequence technology gives its greatest benefit is medicine — in particular, cancer treatment.

When Steve Jobs was diagnosed with pancreatic cancer, he paid around \$100,000 to have his genome sequenced in an effort to gain any information he could about his cancer, and possible treatments for it. Unfortunately there wasn't a whole lot that could be done with that information even just a few years ago. Today however, depending on the type of cancer you have, everything from the invasiveness of your particular brand of tumor, its response to any of a host of drugs in your treatment cocktail, and even how your body metabolizes, excretes, or otherwise unravels in allergic response to those

drugs, can be assessed. Regrettably, at the moment at least, the tests for those various factors are still given piecemeal, kind of like paying \$32 to read a journal article for 24 hours when a year's subscription is only \$120. With the cost of sequencing a genome shrinking to \$1000 or less, it will soon make sense for everyone to subscribe.

Nowhere has genetic analysis of cancer patients become more of an issue than for breast cancer. This week the courts began deliberations on the right of one company, Myriad Genetics, to patent your genes — in particular a gene known as BRCA-1. A better way to put that may be to say that the



courts began deliberations on your right to pursue genetically-informed treatments in a timely and affordable manner without the now present monopoly on information that should be in the public domain, regardless of efforts spent to obtain it. It is absurd to live under a system that says just because someone spends resource to solve a problem you happen to have, you have no right to seek relief from any other source. Right now, if

woman wants to know if she has mutations in two common breast cancer genes, she has little or no choice of where she can be tested, for her own gene. Myriad controls that, and virtually the entire market with it.

Immediate issues aside, we have a long way to go towards learning the function of all the genes we have now sequenced. Even more important than knowing these functions, is knowing how all of the genome is organized in the nucleus to define each of our cells, and by implication, our entire organism. The operational structure of nucleus is what we want to understand now.

Beyond that, having a fully-clickable genome, not only for the fertilized egg from which we were conceived, but from any important cell in our bodies at the time it was sequenced, would be invaluable towards understanding our current health and future prognosis. Each major hit of radiation, or of any of a host of other natural insults we receive in the course of life, leaves its mark. Depending on the health of repair mechanisms, faults might be repaired, or they can slip away unnoticed to cause problems at a later date.

The legacy of the understanding of our genomes has just begun. As more people arm themselves with knowledge of their genes, this information will continue to grow in collective value. To share this information openly with others and learn from them in response is a privilege. To fear sharing that knowledge on the weak pretense of abuse will be seen, looking back, not only as cowardice, but ingratitude.